CHAPTER 12

# Parametric representations and Cartesian equations

S65, S67, S70

> **Key point · Objectives.** Describe a line and a plane; determine intersections; choose a method for a projection.

## Course notes

Throughout this chapter, coordinates are given in an orthonormal frame of space. This assumption is necessary to identify a normal vector from the coefficients of an equation and to calculate distances.

### Parametrised lines

The line through $A(x_A, y_A, z_A)$ with non-zero direction vector $(a, b, c)$ has the representation

$$d : \begin{cases} x = x_A + at \\ y = y_A + bt \\ z = z_A + ct \end{cases} \qquad (t \in \mathbb{R}).$$

A point belongs to the line if there exists *one and the same* real number $t$ satisfying the three equalities. A line has many representations: change the starting point or multiply the direction vector by a non-zero real number.

**Example.** $(x, y, z) = (1, 2, 0) + t(2, -1, 3)$ describes the same line as $(3, 1, 3) + s(4, -2, 6)$; the link is $t = 1 + 2s$.

### Cartesian planes

> **Key point · Equation of a plane.** If $(a, b, c) \neq (0, 0, 0)$, the set $ax + by + cz + d = 0$ is a plane with normal vector $(a, b, c)$. The plane with normal $\vec{n}$ through $A$ is constructed by writing $\overrightarrow{AM} \cdot \vec{n} = 0$.

**Proof.** The condition $\overrightarrow{AM} \cdot \vec{n} = 0$ is written $a(x - x_A) + b(y - y_A) + c(z - z_A) = 0$. Expanding, $d = -ax_A - by_A - cz_A$. Conversely, every point satisfying this equation satisfies the orthogonality condition, hence belongs to the plane.

**Example.** The plane with normal $(2, -1, 3)$ through $A(1, 2, 0)$ is $2(x - 1) - (y - 2) + 3z = 0$, that is $2x - y + 3z = 0$.

## Intersections

For a line and a plane, replace $x, y, z$ in the equation of the plane by their parametric expressions. A solution in $t$ gives a single intersection point; an identity gives a line contained in the plane; a contradiction gives a parallel line that is disjoint from the plane. For two planes, non-collinear normals give an intersection along a line; solve the system of two equations by choosing one variable as a parameter. If the normals are collinear, the planes are parallel: distinguish coincident from disjoint. For two lines, use independent parameters $s, t$ and solve the three equations. With no solution and non-collinear direction vectors, the lines are not coplanar.

## Projection onto a plane

If $\mathcal{P} : ax + by + cz + d = 0$, look for the orthogonal projection of $M$ onto the normal line through $M$: $H = M + t(a, b, c)$. Substitute into the equation of the plane to determine $t$, then calculate $MH$. We deduce, without making it a formula to be learned in isolation,

$$d(M, \mathcal{P}) = \frac{|ax_M + by_M + cz_M + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

**Essential check.** After calculating a projection, check simultaneously that $H \in \mathcal{P}$ and that $\overrightarrow{MH}$ is collinear with the normal.

## Research activity

**Three lines facing a plane — 30 min.**

In an orthonormal frame of space, consider the plane $\mathcal{P}$ with equation $x + y + z = 3$. In each question, the parameter $t$ ranges over $\mathbb{R}$.

1. The line $d_1$ has the representation $(x, y, z) = (t, t, t)$. Determine $d_1 \cap \mathcal{P}$.
2. The line $d_2$ has the representation $(x, y, z) = (3 + t, -t, 0)$. Determine $d_2 \cap \mathcal{P}$.
3. The line $d_3$ has the representation $(x, y, z) = (t, -t, 0)$. Determine $d_3 \cap \mathcal{P}$.
4. For each line, state whether the equation obtained after substitution into that of the plane has a unique solution, no solution, or every real number as a solution. Link these cases to the geometric positions found.

## Progressive exercises

**E1 · Practice.** In a frame of space, consider $A(1,-1,2)$ and $B(3,0,1)$. Give a parametric representation of the line $(AB)$.

**E2 · Practice.** In an orthonormal frame, determine a Cartesian equation of the plane through $A(1,2,3)$ with normal vector $\vec{n} = (1,0,-2)$.

**E3 · Practice.** In a frame of space, the line $d$ is given by $(x,y,z) = (t, 1+t, 2-t)$, $t \in \mathbb{R}$, and the plane $\mathcal{P}$ by $x + 2y + z = 8$. Determine their intersection.

**E4 · Practice.** In a frame of space, determine the intersection of the planes with equations $x + y + z = 1$ and $x - y = 0$. Give a parametric representation if the intersection is a line.

**E5 · Practice.** In a frame of space, study the relative position of the planes with equations $2x - y + z = 3$ and $4x - 2y + 2z = 5$. Justify.

## Textbook or past-paper work

**Baccalauréat, Metropolitan France, 17 June 2025: exercise 3, statements 1 to 4 (p. 5–6).** At home after S70. Justify each answer; for a false statement, explain precisely what refutes it. The paper is available here: official past paper. Use the data and instructions of this past paper.

## Homework

**Homework 12 — Projection and intersection — 55 min.**

In an orthonormal frame of space, consider the point $M(3,0,0)$ and the planes $\mathcal{P} : x + 2y + 2z = 9$ and $\mathcal{Q} : x - y = 0$.

1. Give the coordinates of a non-zero normal vector to $\mathcal{P}$.
2. Determine the coordinates of the orthogonal projection $H$ of $M$ onto $\mathcal{P}$, then calculate the distance from $M$ to $\mathcal{P}$.
3. Show that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it.
4. Does the point $H$ belong to this line? Justify.
5. Determine the coordinates of the point $K$ of this line whose third coordinate (the "cote") is zero. Calculate $MK$, then compare $MK$ and $MH$.